

Chen Hajaj - March 2026

CONTACT INFORMATION	Department of Industrial Engineering and Management https://www.ariel.ac.il/wp/chen-hajaj/	Ariel University chenha@ariel.ac.il
RESEARCH INTERESTS	Primary Areas: Machine Learning, Medical Data Science, Cybersecurity, Encrypted Traffic Classification Applications: E-commerce, Network Security, Clinical Decision Support, Healthcare Analytics Theoretical: Behavioral Game Theory, Crowdsourcing, Multi-Agent Systems, Artificial Intelligence	
ACADEMIC POSITIONS	Industrial Engineering and Management, Ariel University, Israel Associate Professor 2025 – present Senior Lecturer (Tenured) 2022 – 2025 Lecturer 2018 – 2022	
LEADERSHIP AND SERVICE ROLES	University Ambassador, NVIDIA 2025 – present Representing academic initiatives and promoting GPU computing Director, Data Science Research Center 2018 – 2024 Led interdisciplinary data science research initiatives and established collaborations Chair, Faculty of Engineering Review Board 2022 – present Overseeing academic standards and evaluation processes for engineering programs Data Science Fellow, Vanderbilt University 2017 – 2018 Data think tank - Tennessee Educational Research Alliance (TERA) Data think tank - Anthropogenic landscape features in South America Postdoc Researcher 2016 – 2018 Electrical Engineering and Computer Science, Vanderbilt University Host: Vorobeychik Yevgeniy, Ph.D.	
EDUCATION	Bar-Ilan University , Ramat Gan, Israel Ph.D., Computer Science 2016 Thesis Topic: <i>Intelligent Mechanisms for Platforms in Modern Markets</i> Advisors: Sarne David, Ph.D. and Hassidim Avinatan, Ph.D. M.Sc., Electrical Engineering 2012 Topic: <i>Three-Dimensional Group Registration of Mesh Objects</i> Advisor: Keller Yosi, Ph.D. B.Sc., Computer Engineering 2010	
REFEREED JOURNAL PUBLICATIONS	1. Chen Hajaj , Shani Alkoby, Shai Ashkenazi, Avner Herman Cohen, Yael Reichenberg, Shiri Kushnir, Vered Shkalim Zemer. Machine Learning Tools for Predicting Pediatric Urinary Tract Infections Caused by ESBL-producing Bacteria. <i>The Pediatric Infectious Disease Journal</i> . 2026	

2. Zohar Simhon, Matan Weiss, Revital Marbel, **Chen Hajaj**, Amit Dvir, Ran Dubin. Real-Time Network Security: Integrating ANN and Dynamic Graph-Based Clustering. *Computer Networks*. 2026
3. Ron Hirshprung, Ori Lesman, **Chen Hajaj**. The Sound of Emotions, An Artificial Intelligence Approach to Predicting Emotions from Musical Selections. *Multimedia Systems*. 2026
4. Anat Goldstein, **Chen Hajaj**, Pierre Singer, Orit Raphaeli. Survival Forest Models for ICU Mortality Prediction based on Nutrition and Clinical Factors. *IEEE Access*. 2026
5. Zafrir Avrahami, Moti Zwilling, **Chen Hajaj**. Leveraging OSINT for Advanced Proactive Cybersecurity: Strategies and Solutions. *IEEE Access*. 2025
6. Ro'i Lang, Or Haim Anidjar, Sahar Slonimsky, **Chen Hajaj**, Oz Golan, Carmel Matias, Alex Diskin, Evgeny Stokin, Mor Mega. Measuring and Analyzing Defects of Additive Manufactured Ti-6Al-4V Specimens Through Image Segmentation. *Fatigue & Fracture of Engineering Materials & Structures*. 2025.
7. Udi Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. A classification-by-retrieval framework for few-shot anomaly detection to detect API injection. *Computers & Security*. 2025
8. Or Haim Anidjar, Revital Marbel, Ran Dubin, Amit Dvir, **Chen Hajaj**. Extending Limited Datasets with GAN-Like Self-Supervision for SMS Spam Detection. *Computers & Security*. 2024
9. Assaf Frajman, Shimon Izhakian, Ori Mekiten, Ori Hadar, Ariel Lichtenstadt, **Chen Hajaj**, Shon Schori, Moshe Heching, Dror Rosengarten and Mordechai R. Kramer. Phenotypical Characteristics of Nontuberculous Mycobacterial Infection in Patients with Bronchiectasis. In *Respiratory Research*. 2024
10. **Chen Hajaj**, Porat Aharon, Ran Dubin, and Amit Dvir. The Art of Time-Bending: Data Augmentation and Early Prediction for Efficient Traffic Classification. In *Expert Systems with Applications*. 2024
11. Anat Goldstein and **Chen Hajaj**. Measuring flight-destination similarity: A multidimensional approach. In *Expert Systems with Applications*. 2024
12. Ofek Bader, Adi Lichy, Amit Dvir, Ran Dubin, and **Chen Hajaj**. OSF-EIMTC: An Open-Source Framework for Standardized Encrypted Internet Traffic Classification. *Computers Communications*. 2024
13. Orit Raphaeli, Liran Statlender, **Chen Hajaj**, Itai Bendavid, Anat Goldstein, Eyal Robinson, and Pierre Singer. Using Machine-Learning to Assess the Prognostic Value of Early Enteral Feeding Intolerance in Critically Ill Patients: A Retrospective Study. *Nutrients*. 2023
14. Harel Berger, Amit Dvir, Enrico Mariconti, and **Chen Hajaj**. Breaking the structure of MaMaDroid. *Expert Systems with Applications*. 2023

15. Ofir Joseph, Avshalom Elmalech, and **Chen Hajaj**. Detecting Parallel Covert Data Transmission Channels in Video. *Electronics*. 2023
16. Or Haim Anidjar, Yannick Estève, **Chen Hajaj**, Amit Dvir, Itshak Lapidot. Speech and Multilingual Natural Language Framework for Speaker Change Detection and Diarization. *Expert Systems With Applications*. 2023
17. Adi Lichy, Ofek Bader, Ran Dubin, Amit Dvir, **Chen Hajaj**. When a RF Beats a CNN and GRU, Together - A Comparison of Deep Learning and Classical Machine Learning Approaches for Encrypted Malware Traffic Classification. *Computers & Security*. 2023
18. Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich, Corine Berger. The Online temporary learning groups in higher education: Interactions, compensation, and maximization of achievements in an Israeli case study. *Journal of Education Culture and Society*. 2023
19. Rotem Bar and **Chen Hajaj**. SimCSE for Encrypted Traffic Detection and Zero-Day Attack Detection. *IEEE ACCESS*. 2022
20. **Chen Hajaj**, Nitay Hason, and Amit Dvir. Less is More: Robust and Novel Features for Malicious Domain Detection. *Electronics*. 2022
21. Anat Goldstein and **Chen Hajaj**. The Hidden Conversion Funnel of Mobile vs. Desktop Consumers. *Electronic Commerce Research and Applications (ECRA)*. 2022
22. Corine Berger, Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich. Patterns of Activity in Temporary Digital Learning Groups, and Its Correlation With Academic Achievement Among University Students in an Israeli University. *Social Issues in Israel*. 2022
23. Harel Berger, **Chen Hajaj**, and Amit Dvir. Crystal Ball: From Innovative Attacks to Attack Effectiveness Classifier. *IEEE ACCESS*. 2021
24. Ron Hirschprung and **Chen Hajaj**. Prediction Model for the Spread of the COVID-19 Outbreak in the Global Environment. *Heliyon*. 2021
25. Or Anidjar, Itshak Lapidot, **Chen Hajaj**, Amit Dvir and Issachar Gilad. Hybrid Speech and Text Analysis Methods for Speaker Change Detection. *IEEE/ACM Transactions on Audio Speech and Language Processing*. 2021
26. **Chen Hajaj**, Zlatko Joveski, Sixie Yu, Yevgeniy Vorobeychik. Robust Coordination in Adversarial Social Networks: From Human Behavior to Agent-Based Modeling. *Network Science*. 2021
27. Amit Dvir, Angelos K. Marnerides, Ran Dubin, Nehor Golan, and **Chen Hajaj**. Encrypted Video Traffic Clustering Demystified. *Computers & Security*, 96:101854, 2020
28. **Chen Hajaj** and David Sarne. Selective Opportunity Disclosure at the Service of Strategic Information Platforms. *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(5):1133–1164, 2017

29. **Chen Hajaj**, Noam Hazon, and David Sarne. Enhancing Comparison Shopping Agents through Ordering and Gradual Information Disclosure. *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(3):696–714, 2017
30. **Chen Hajaj**, Noam Hazon, and David Sarne. Improving Comparison Shopping Agents’ Competence through Selective Price Disclosure. *Electronic Commerce Research, and Applications (ECRA)*, 14(6):563–581, 2015

REFEREED
CONFERENCE
PUBLICATIONS

1. Udi Aharon, Amit Dvir, Ran Dubin, Revital Marbel, **Chen Hajaj**. Uncovering Microservice Faults: A Temporal Graph Approach to Root Cause Analysis. In *IEEE International Conference on Communications (ICC 2026)*, Glasgow, Scotland, UK, May 2026.
2. Meir Litman and **Chen Hajaj**. Cleaner Adversarial CAPTCHAs: Intelligent Targets and Precise Noise for Usable Security. In *The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2026
3. Yehonatan Zion, Eyal Paz, Ran Dubin, Amit Dvir, **Chen Hajaj**. Quality of Experience Prediction for First Person Shooter Online Gaming: The Case Study of Call of Duty. In *IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, January 2026.
4. Revital Marbel, Yanir Victor Cohen, Ran Dubin, Amit Dvir, **Chen Hajaj**. Cloudy with a Chance of Anomalies: Dynamic Graph Neural Network for Early Detection of Cloud Services’ User Anomalies. In *IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, January 2026.
5. Simona Lisker, Ayelet Botman, **Chen Hajaj**, Ran Dubin, Amit Dvir. Optimized File Type Detection and One-Shot Retrieval. In *IEEE International Conference on Communications (ICC)*. 2025
6. Zohar Simhon, Matan Weiss, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. A New D-MAGIC: Dynamic Model for Cybersecurity Attack Detection Using GNNs into Clustering. In *IEEE International Conference on Communications (ICC)*. 2025
7. Yehonatan Zion, Porat Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. Enhancing Encrypted Internet Traffic Classification Through Advanced Data Augmentation Techniques. In *IEEE International Conference on Communications (ICC)*. 2025
8. Angelos Marnierides, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. PQClass: Classification of Post-Quantum Encryption Applications in Internet Traffic. In *IEEE International Conference on Communications (ICC)*. 2025
9. Anat Goldstein and **Chen Hajaj**. Warm Recommendation: Enhancing Cold Start Recommendations Using Multimodal Product Representations. In *International conference of the Association for Information Systems (ICIS)*. 2024

10. Nathan Dillbary, Roi Yozevitch, Amit Dvir, Ran Dubin, and **Chen Hajaj**. Hidden in Time, Revealed in Frequency: Spectral Features and Multiresolution Analysis for Encrypted Internet Traffic Classification. In *IEEE Consumer Communications & Networking Conference (CCNC)*. 2024
11. Anat Goldstein and **Chen Hajaj**. Warming Up the Cold Start: A Multimodal Approach. In *Special Interest Group on Decision Support and Analytics (SIGDSA) Symposium*. 2023
12. Orit Raphaeli, Liran Statlander, **Chen Hajaj**, Itai Ben David, and Pierre Singer. Protein Intake and Clinical Outcomes of Critically Ill Patients. In *the 44th European Congress for Clinical Nutrition and Metabolism (ESPEN)*. 2022
13. Anat Goldstein and **Chen Hajaj**. Measuring Product Similarity: A Multidimensional Approach. In *the Eighteenth Symposium on Statistical Challenges in Electronic Commerce Research*. 2022
14. Ofek Bader, Adi Lichy, **Chen Hajaj**, Ran Dubin, and Amit Dvir. MalDIST: From Encrypted Traffic Classification to Malware Traffic Detection and Classification. In *IEEE Consumer Communications & Networking Conference (CCNC)*. 2022
15. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer, and Uri Gabbay. Using the Cardio-Vascular Index (CVRI) To Predict Mortality in Septic Shock. In *the 40th International Symposium on Intensive Care & Emergency Medicine*. 2022
16. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Support Early Prediction of Feeding Intolerance in Critically Ill Patients. In *Intensive Care Medicine Experimental*. 2021
17. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Compare Gastric Residual Volume Thresholds as Predictors of Clinical Outcomes in Critically Ill Patients. In *International Symposium of Intensive Care and Emergency Medicine*. 2021
18. Or Anidjar, Itshak Lapidot, **Chen Hajaj** and Amit Dvir. A Thousand Words are Worth More Than One Recording: Word-Embedding Based Speaker Change Detection. In *Interspeech*. 2021
19. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Feeding Intolerance as a Predictor of Clinical Outcomes in Critically Ill Patients: A Machine Learning Approach. In *the 43rd European Society for Clinical Nutrition and Metabolism (ESPEN)*. 2021
20. Uzi Ben Shalom, **Chen Hajaj**, Nitza Davidovich and Corinne Berger. Online temporary learning groups in higher education – Interactions, compensation, and maximization of achievements in an Israeli case-study. In *the 1st international e-conference on teacher education generation next: perspectives, opportunities & challenges*. 2021

21. Vitali Demianiuk, **Chen Hajaj**, and Kiril Kogan. PCL: Packet Classification with Limited Knowledge. *INFOCOM*. 2021
22. Anat Goldstein and **Chen Hajaj**. The Different Path to Purchase of Mobile and Desktop Consumers: Analyzing Consumers' Progress in the Conversion Funnel Using Hidden Markov Models. In *Statistical Conference in E-Commerce Research (SCECR)*. 2020
23. Nitay Hason, Amit Dvir, and **Chen Hajaj**. Robust Malicious URL Detection. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML)*. 2020
24. Liang Tong, Bo Li, **Chen Hajaj**, Chaowei Xiao, Ning Zhang, Yevgeniy Vorobeychik. Improving Robustness of ML Classifiers against Realizable, Evasion Attacks Using Conserved Feature. In the *28th USENIX Security Symposium*. 2019
25. Yevgeniy Vorobeychik, Myrna Wooders, Greg Leo, Matthew Chambers, **Chen Hajaj**, Jian Lou, and Martin Van der Linden. Non-Cooperative Team Formation and a Team Formation Mechanism. In the *11th annual conference of the Israeli chapter of the game theory society*. 2019
26. **Chen Hajaj**, Sixie Yu, Zlatko Joveski, Yifan Guo, and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2019
27. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin, and Daniel Fabbri. Crowdsourcing Clinical Chart Reviews. In *AMIA Joint Summits on Translational Science*. 2018
28. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial task assignment. In *International Joint Conference on Artificial Intelligence and European Conference on Artificial Intelligence (IJCAI-ECAI)*. 2018
29. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin and Daniel Fabbri. A Crowdsourcing Framework for Medical Data Sets. In *American Medical Informatics Association (AMIA)*. 2018
30. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Enhancing Crowdworkers' Vigilance. In *International Joint Conference on Artificial Intelligence (IJCAI)*. 2017
31. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Extending Workers' Attention Span Through Dummy Sub-Goals. In *AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*. 2016. **Best paper finalist.**
32. **Chen Hajaj**, John P. Dickerson, Avinatan Hassidim, Tuomas Sandholm and David Sarne. Strategy-Proof and Efficient Kidney Exchange Using a Credit Mechanism. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2015

33. Tal Darom, Yaniv Gur, **Chen Hajaj** and Yosi Keller. Automatic Anatomical Shape Correspondence and Alignment Using Mesh Features. In *International Symposium on Biomedical Imaging (ISBI)*. 2015
34. **Chen Hajaj** and David Sarne. Strategic Information Platforms - Selective Disclosure and The Price of "Free". In *ACM Conference on Economics and Computation (EC)*. 2014
35. **Chen Hajaj**, Noam Hazon, and David Sarne. Ordering Effects and Belief Adjustment in the Use of Comparison Shopping Agents. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2014
36. **Chen Hajaj**, Noam Hazon, David Sarne, and Avshalom Elmalech. Search More, Disclose Less. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2013

SHORT PAPERS
AND REFEREED
WORKSHOPS

1. Brigitta R Schvarcz and **Chen Hajaj**. Exploring Student Perspectives of the Impact of VR in ESP. In *TESOL*. 2024
2. Bar Vinder, **Chen Hajaj**, and Idit Avrahami. A method for evaluating the performance of a novel percutaneous Left Ventricular Assist Device and optimizing its design. In *VPH 2022 PHYSIOLOGICAL HUMAN CONFERENCE*. 2022
3. Harel Berger, **Chen Hajaj**, and Amit Dvir. Evasion is not enough: A case study of Android malware. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML)*. 2020
4. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial Randomized Task Assignment, 6th Day on Computational Game Theory. 2019
5. **Chen Hajaj**, Sixie Yu, Zlatko Joveski ,and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence(AI³)*. 2018
6. Jian Lou, **Chen Hajaj**, Myrna Wooders and Yevgeniy Vorobeychik. Automated Mechanism Design for the Roommates Problem. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence (AI³)*. 2018
7. Jian Lou, Martin Van der Linden, Pranav Batra, **Chen Hajaj**, Greg Leo, Yevgeniy Vorobeychik and Myrna Wooders. Rotating Proposer Mechanisms for Team Formation. In *Workshop on Cooperative Games in Multiagent Systems (CoopMAS)*.2017. **Most visionary paper.**
8. **Chen Hajaj**, David Sarne, and Lea Perets. Advanced Service Schemes for a Self-Interested Information Platform. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2014

BOOK CHAPTERS

Sofia Ceppi, Esther David, **Chen Hajaj**, Valentin Robu, and Ioannis A. Vetsikas, editors, Agent-Mediated Electronic Commerce: Designing Trading Strategies and Mechanisms for Electronic Markets, Springer, LNBIP 271, 2017.

PATENTS

- Eyal PAZ, Ran Dubin, **Chen HAJAJ**, and Amit DVIR, "first-person shooter online gaming QOE prediction from network traffic analysis", **US provisional application**
- Amir Glik, **Chen Hajaj**, Orit Rephaeli, Anat Goldstein. "PREDICTING ONSET OF NEURODEGENERATIVE DISEASES" **US Provisional Patent Application No. 63/456,555**
- **Chen Hajaj**, David Sarne, Noam Hazon. "Selective Options Disclosure Method for Affecting a User's Decision-Making" **US Provisional Patent Application No. 61/842,974**

STUDENTS

Ph.D. students

1. Ohad Nir, Ariel University	2025 – present
2. Tanya Zhelezni, Ariel University	2025 – present
3. Zohar Shimhon, Ariel University	2025 – present
4. Shira Schweitzer, Ariel University	2025 – present
5. Bar Vinder, Ariel University	2025 – present
6. Omri Arbiv, Ariel University	2025 – present
7. Natan Dilbberry, Ariel University	2024 – present
8. Aviad Levi, Ariel University	2023 – present
9. Udi Aharon, Ariel University	2022 – present
10. Yehonatan Zion, Ariel University	2022 – present
11. Harel Berger, Ariel University	2019 – 2022
12. Or Anidjar, Ariel University	2019 – 2022

Pre-Ph.D.

1. Liav Cohen, Ariel University	2026 – present
2. Zafrir Avrahami, Ariel University	2023 – 2024
3. Omri Arbiv, Ariel University	2022 – 2024
4. Meir Litman, Ariel University	2020 – 2023

Masters (Thesis)

1. Idan Buzaglo, Ariel University	2025 – present
2. Tuvia Smadar, Ariel University	2025 – present
3. Ihab Hino, Ariel University	2025 – present
4. Amit Stein, Ariel University	2025 – present
5. Eylon Yehiel, Ariel University	2024 – present
6. Yona Elbar, Ariel University	2024 – present
7. Porat Aharon, Ariel University	2022 – 2023
8. Ofir Yosef, Bar-Ilan University	2021 – 2023
9. Bar Vinder, Ariel University	2021 – 2023
10. Adi Lichi, Ariel University	2021 – 2023
11. Ofek Bader, Ariel University	2021 – 2022
12. Nitay Hason, Ariel University	2019 – 2020
13. Nabeeha Yassin, Ben-Gurion University	2018 – 2021

Masters (Final project)

1. Elad Chen, Ariel University	2021 – 2023
2. Tal Abed, Ariel University	2021 – 2022
3. Ariel Yehezkel, Ariel University	2021 – 2022
4. Bar Ben David, Afeka College	2022 – 2024
5. Ben Ron Udi, Afeka College	2022 – 2024
6. Chen Shiener, Afeka College	2022 – 2023
7. Nathanael Guinzburg, Afeka College	2022 – 2023
8. Michael Gudovsky, Afeka College	2022 – 2023
9. Rotem Bar, Afeka College	2021 – 2022
10. Pavel Shochet, Afeka College	2021 – 2022
11. Eliran Barai, Afeka College	2020 – 2021

Program Committee

ILAIS, ICIS, AAAI, AAMAS, ACM EC, AMIA, AMEC, CPSCCom, GameSec, ICAART, IJCAI, SODA, ECAI.

Reviewer

Annals of Mathematics and Artificial Intelligence, Journal of Information Security and Applications (JISA), Artificial Intelligence Review (AIRE), Journal of Artificial Intelligence Research (JAIR), Computers & Security, Expert Systems with Applications, Computer Networks, Health Systems, Transactions on Knowledge and Data Engineering.